Introduction to Automated Testing

This is the first page of Sample Document 1, specifically created for the iLovePDF Selenium TestNG automation framework. It is used as the primary upload file for Merge PDF, Split PDF, Compress PDF, PDF to Word, PDF to PowerPoint, and PDF to Excel test scenarios.

Automated testing ensures that software behaves correctly under a wide variety of conditions. By combining Page Object Model (POM) with TestNG data providers, this framework covers positive, negative, boundary, and end-to-end test scenarios in a clean, maintainable structure.

This document was generated programmatically using ReportLab and is intentionally formatted to include headings, paragraphs, and a data table on each page so that conversion tools have meaningful content to process.

Data Table

TC ID	Feature	Type	Group
TC_MERGE_001	Merge PDF	Positive	Regression
TC_SPLIT_001	Split PDF	Positive	Regression
TC_COMPRESS_001	Compress PDF	Positive	Regression
TC_PDF2W_001	PDF to Word	Positive	Regression
TC_E2E_001	Merge PDF	E2E	E2E

Test Architecture Overview

The iLovePDF automation framework is built on a layered architecture. At the lowest level, DriverFactory and DriverManager manage the WebDriver lifecycle using ThreadLocal for parallel execution safety.

Helper classes (ClickHelper, InputHelper, ElementHelper, ValidationHelper, FileHelper) wrap raw Selenium APIs with explicit waits, reducing flakiness and improving readability. All page interactions in the Page Object layer delegate to these helpers through BasePage.

Test data is externalised into Excel sheets (via Apache POI) and JSON files (via Jackson) and fed to tests through TestNG DataProviders. This ensures test logic and test data are fully decoupled.

Data Table

Layer	Class	Responsibility	Pattern
Driver	DriverFactory	Browser init	Factory
Driver	DriverManager	ThreadLocal storage	Singleton
Page	BasePage	Common actions	POM Base
Test	BaseTest	Setup / teardown	Template Method
Utility	WaitUtil	Synchronisation	Wrapper

Test Execution Summary – Page 3

This third page ensures that Split PDF tests have multiple pages to work with, and that Merge PDF tests produce a meaningful output when two or three documents are combined. A three-page document is the minimum recommended size for thorough split testing.

All test cases are tagged with TestNG groups (smoke, regression, e2e) and are controlled by separate TestNG suite XML files. The RetryAnalyzer listener automatically retries failed tests up to the configured retry count, reducing false negatives caused by transient network issues.

ExtentReports generates a timestamped HTML report for every suite run, with automatic screenshot attachment on failure via the TestListener. Log4j2 provides structured console and file logging throughout execution.

Data Table

Suite	XML File	Groups	Tests
Smoke	smoke.xml	smoke	14
Regression	regression.xml	regression	99
E2E	e2e.xml	e2e	16
Sanity	sanity.xml	selected	5
TOTAL	—	all	134